

COMPARATIVE ANALYSIS

Evaluation and Performance Metrics of Object Detection, Motion Estimation, and Pattern Recognition Techniques

1. Detection Performance Comparison (Viola-Jones vs. YOLO)

Given Data: Model A (Viola-Jones): TP = 80, FP = 20, FN = 30 | Model B (YOLO): TP = 90, FP = 15, FN = 20

(a) Calculations:

- Formulae:** $Precision = TP / (TP + FP)$ | $Recall = TP / (TP + FN)$
- Model A (Viola-Jones):** Precision = $80 / (80 + 20) = 80 / 100 = 0.8000$ (80.00%) | Recall = $80 / (80 + 30) = 80 / 110 = 0.7273$ (72.73%)
- Model B (YOLO):** Precision = $90 / (90 + 15) = 90 / 105 = 0.8571$ (85.71%) | Recall = $90 / (90 + 20) = 90 / 110 = 0.8182$ (81.82%)

(b) Comparison & Justification: Model B (YOLO) is significantly better. It achieves higher Precision (85.71% vs. 80.00%) and higher Recall (81.82% vs. 72.73%), demonstrating both lower false alarm rates and superior sensitivity in detecting targets compared to the traditional Viola-Jones framework.

2. Motion Estimation Error Analysis

Given Data: Displacement errors in pixels across 3 frames — Method A: [2, 3, 2] | Method B: [3, 5, 4]

(a) Calculations:

- Formula:** $Average\ Error = (\sum Errors) / N$
- Method A:** Average Error = $(2 + 3 + 2) / 3 = 7 / 3 = 2.3333$ pixels
- Method B:** Average Error = $(3 + 5 + 4) / 3 = 12 / 3 = 4.0000$ pixels

(b) Comparison & Justification: Method A is substantially more accurate. Method A yields a lower mean displacement error of 2.33 pixels compared to Method B's 4.00 pixels, indicating tighter alignment with real pixel movement across frame sequences.

3. Optical Flow Magnitude Comparison

Given Data: Optical flow magnitudes (pixels/frame) across regions R1, R2, R3 — Method A: [5, 6, 5] | Method B: [8, 10, 9]

(a) Calculations:

- Formula:** $Average\ Magnitude = (\sum Magnitudes) / N$
- Method A:** Average Magnitude = $(5 + 6 + 5) / 3 = 16 / 3 = 5.3333$ pixels/frame
- Method B:** Average Magnitude = $(8 + 10 + 9) / 3 = 27 / 3 = 9.0000$ pixels/frame

(b) Comparison & Justification: Method B captures significantly stronger motion. With an average magnitude of 9.00 pixels/frame versus Method A's 5.33 pixels/frame, Method B exhibits higher responsiveness to larger vector displacements in visual flow fields.

4. Depth Estimation Comparison (Stereo Vision vs. Structure from Motion)

Given Data: Actual depths — A = 2.0m, B = 3.0m, C = 4.0m.

Stereo Vision estimates: [2.0, 3.0, 4.0] | Structure from Motion (SfM) estimates: [2.5, 3.8, 5.0]

(a) Calculations:

- **Formula:** $Absolute\ Error = |Estimated - Actual|$
- **Stereo Vision Errors:** A: $|2.0 - 2.0| = 0.0m$ | B: $|3.0 - 3.0| = 0.0m$ | C: $|4.0 - 4.0| = 0.0m$ → **Mean Absolute Error (MAE) = 0.00m**
- **Structure from Motion (SfM) Errors:** A: $|2.5 - 2.0| = 0.5m$ | B: $|3.8 - 3.0| = 0.8m$ | C: $|5.0 - 4.0| = 1.0m$ → **Mean Absolute Error (MAE) = $(0.5 + 0.8 + 1.0) / 3 = 0.7667m$**

(b) Comparison & Justification: Stereo Vision is perfectly accurate in this benchmark, achieving zero absolute error across all target objects, whereas SfM exhibits a systematic overestimation bias leading to an average error of 0.77 meters.

5. Pattern Recognition Accuracy Comparison

Given Data: Traditional System: 85 correct out of 100 | Deep Learning System: 92 correct out of 100

(a) Calculations:

- **Formula:** $Accuracy = (Correct\ Predictions / Total\ Samples) \times 100\%$
- **Traditional System:** $Accuracy = (85 / 100) \times 100\% = 85.00\%$
- **Deep Learning System:** $Accuracy = (92 / 100) \times 100\% = 92.00\%$

(b) Comparison & Justification: Deep Learning performs superiorly. Outperforming the traditional system by 7 percentage points (92% vs. 85%), deep learning feature abstractions capture complex data patterns more effectively.

6. Precision-Recall Trade-off Analysis

Given Data: Model A: TP = 70, FP = 30, FN = 20 | Model B: TP = 85, FP = 25, FN = 15

(a) Calculations:

- **Model A:** Precision = $70 / (70 + 30) = 0.7000$ (70.00%) | Recall = $70 / (70 + 20) = 0.7778$ (77.78%)
- **Model B:** Precision = $85 / (85 + 25) = 0.7727$ (77.27%) | Recall = $85 / (85 + 15) = 0.8500$ (85.00%)

(b) Comparison & Justification: Model B is significantly better balanced. Model B achieves both a higher precision (77.27% vs 70.00%) and higher recall (85.00% vs 77.78%), overcoming the trade-off limitations observed in Model A.

7. F1-Score Comparison

Given Data: Model A: Precision = 0.80, Recall = 0.60 | Model B: Precision = 0.75, Recall = 0.75

(a) Calculations:

- **Formula:** $F1\text{-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
- **Model A:** $F1 = 2 \times (0.80 \times 0.60) / (0.80 + 0.60) = 0.96 / 1.40 = 0.6857 \text{ (68.57\%)}$
- **Model B:** $F1 = 2 \times (0.75 \times 0.75) / (0.75 + 0.75) = 1.125 / 1.50 = 0.7500 \text{ (75.00\%)}$

(b) Comparison & Justification: **Model B is overall superior.** F1-score is the harmonic mean of precision and recall. Model B's perfectly balanced metrics yield an F1-score of 0.7500 versus 0.6857 for Model A, which suffers due to a lower recall.

8. Optical Flow Consistency

Given Data: Motion values across Frames 1, 2, 3 — Method A: [4, 5, 6] | Method B: [6, 8, 9]

(a) Calculations:

- **Average Motion:** Method A = $(4 + 5 + 6) / 3 = 5.0000$ | Method B = $(6 + 8 + 9) / 3 = 7.6667$
- **Sample Variance (s^2):**
Method A: $[(4-5)^2 + (5-5)^2 + (6-5)^2] / 2 = (1 + 0 + 1) / 2 = 1.0000$
Method B: $[(6-7.67)^2 + (8-7.67)^2 + (9-7.67)^2] / 2 = (2.78 + 0.11 + 1.78) / 2 = 2.3333$

(b) Comparison & Justification: **Method A is more stable and consistent.** Method A exhibits lower motion variance (1.00 vs. 2.33), signifying smooth and predictable optical flow trajectories with minimal frame-to-frame jitter.

9. Detection Accuracy Comparison

Given Data: Model A: 180 correct out of 200 | Model B: 170 correct out of 200

(a) Calculations:

- **Model A Accuracy:** $(180 / 200) \times 100\% = 90.00\%$
- **Model B Accuracy:** $(170 / 200) \times 100\% = 85.00\%$

(b) Comparison & Justification: **Model A outperforms Model B** by 5 percentage points in overall object detection accuracy, making it more dependable across testing instances.

10. Error Rate Comparison

Given Data: Method A: 20 errors out of 100 | Method B: 10 errors out of 100

(a) Calculations:

- **Formula:** $\text{Error Rate} = (\text{Errors} / \text{Total}) \times 100\%$
- **Method A:** $(20 / 100) \times 100\% = 20.00\%$
- **Method B:** $(10 / 100) \times 100\% = 10.00\%$

(b) Comparison & Justification: Method B is significantly better. It cuts the error rate in half (10.00% vs 20.00%), ensuring higher statistical reliability and fewer misclassifications.

11. Motion Estimation Variance

Given Data: Frame errors — Frame A: [1, 2, 3] | Frame B: [6, 7, 5]

(a) Calculations:

• Mean Errors:

Method A = $(1 + 2 + 3) / 3 = 2.0000$ pixels

Method B = $(6 + 7 + 5) / 3 = 6.0000$ pixels

• Sample Variance (s^2):

Method A: $[(1-2)^2 + (2-2)^2 + (3-2)^2] / 2 = 1.0000$

Method B: $[(6-6)^2 + (7-6)^2 + (5-6)^2] / 2 = 1.0000$

(b) Comparison & Justification: Method A is overall superior. While both methods demonstrate identical stability and variance ($s^2 = 1.00$), Method A has a significantly lower absolute error magnitude (2.00 pixels vs 6.00 pixels).

12. Intersection Over Union (IoU) Comparison

Given Data: IoU values across Objects 1, 2, 3 — Method A: [0.60, 0.70, 0.65] | Method B: [0.80, 0.85, 0.90]

(a) Calculations:

• Formula: $\text{Mean IoU} (mIoU) = (\sum \text{IoU}) / N$

• **Method A:** $mIoU = (0.60 + 0.70 + 0.65) / 3 = 1.95 / 3 = 0.6500$

• **Method B:** $mIoU = (0.80 + 0.85 + 0.90) / 3 = 2.55 / 3 = 0.8500$

(b) Comparison & Justification: Method B provides significantly better detection quality. An IoU score of 0.85 indicates highly accurate bounding box overlap with ground truth, whereas Method A (0.65) exhibits poorer localization alignment.

13. Recall Comparison in Detection

Given Data: Model A: TP = 60, FN = 40 | Model B: TP = 80, FN = 20

(a) Calculations:

- **Model A Recall:** $60 / (60 + 40) = 60 / 100 = 0.6000$ (60.00%)
- **Model B Recall:** $80 / (80 + 20) = 80 / 100 = 0.8000$ (80.00%)

(b) Comparison & Justification: **Model B displays higher sensitivity.** Achieving 80% recall versus Model A's 60%, Model B successfully identifies 20% more actual targets and reduces missed detections (FN) by half.

14. Depth Estimation Squared Error Analysis

Given Data: Actual depths = [2.0, 3.0, 4.0]m.

Method A Predicted = [2.5, 3.5, 4.5] | Method B Predicted = [2.2, 3.1, 4.2]

(a) Calculations:

- **Formula:** $Squared\ Error = (Predicted - Actual)^2$
- **Method A Squared Errors:** $(2.5-2)^2 = 0.25$ | $(3.5-3)^2 = 0.25$ | $(4.5-4)^2 = 0.25$ → **Mean Squared Error (MSE) = $0.2500\ m^2$**
- **Method B Squared Errors:** $(2.2-2)^2 = 0.04$ | $(3.1-3)^2 = 0.01$ | $(4.2-4)^2 = 0.04$ → **Mean Squared Error (MSE) = $(0.04 + 0.01 + 0.04) / 3 = 0.0300\ m^2$**

(b) Comparison & Justification: **Method B is far more accurate.** Method B's MSE of $0.0300\ m^2$ is over 8 times lower than Method A's MSE of $0.2500\ m^2$, proving significantly tighter depth reconstruction.

15. Recognition Precision Comparison

Given Data: System A: TP = 90, FP = 10 | System B: TP = 85, FP = 5

(a) Calculations:

- **System A Precision:** $90 / (90 + 10) = 90 / 100 = 0.9000$ (90.00%)
- **System B Precision:** $85 / (85 + 5) = 85 / 90 = 0.9444$ (94.44%)

(b) Comparison & Justification: **System B yields higher precision.** System B achieves 94.44% precision compared to System A's 90.00%, producing fewer false positives per positive prediction.